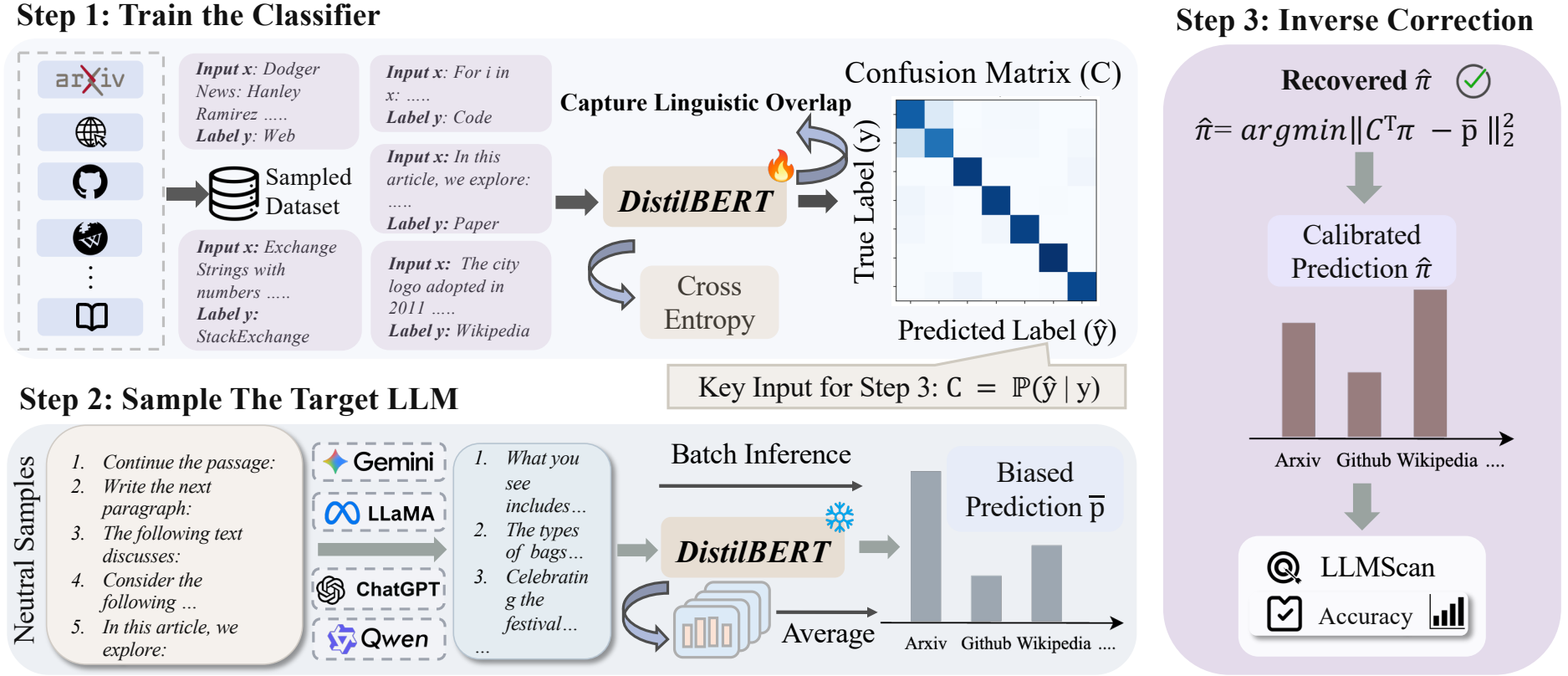
Step 1: Train the Classifier
arXiv
Input x: Dodger News: Hanley Ramirez .....
Label y: Web
Input x: For i in x: .....
Label y: Code
Sampled Dataset
Input x: In this article, we explore: .....
Label y: Paper
Input x: Exchange Strings with numbers .....
Label y: StackExchange
Input x: The city logo adopted in 2011 .....
Label y: Wikipedia
Capture Linguistic Overlap
DistilBERT
Cross Entropy
Confusion Matrix (C)
True Label (y)
Predicted Label (ŷ)
Key Input for Step 3: C = ℙ(ŷ | y)
Step 2: Sample The Target LLM
Neutral Samples
1. Continue the passage:
2. Write the next paragraph:
3. The following text discusses:
4. Consider the following ...
5. In this article, we explore:
Gemini
LLaMA
ChatGPT
Qwen
1. What you see includes...
2. The types of bags...
3. Celebrating the festival...
...
Batch Inference
DistilBERT
Average
Biased Prediction p̄
Arxiv Github Wikipedia ....
Step 3: Inverse Correction
Recovered π̂
π̂= argmin||C^T π − p̄ ||_2^2
Calibrated Prediction π̂
Arxiv Github Wikipedia ....
LLMScan
Accuracy